

SOLUTIONS - Exercise - Week 7: MATH1061/7861

- (1) (4 marks) Consider the sequence defined by $a_1 = 3$, $a_2 = 8$ and

$$a_n = 4(a_{n-1} - a_{n-2})$$

Use Strong Induction to prove $a_n = 2^{n-1}(n+2)$.

Proof:

Base Cases: Observe that

$$a_1 = 3 = 2^0 \times 3 = 2^{1-1}(1+2)$$

and

$$a_2 = 8 = 2^1 \times 4 = 2^{2-1}(2+2).$$

Inductive Hypothesis: Suppose that for some $k \in \mathbb{Z}_{\geq 2}$, for all $n \in \{1, 2, 3, \dots, k\}$ we have $a_n = 2^{n-1}(n+2)$.

Want to Prove: We want to show that $a_{k+1} = 2^{k+1-1}((k+1)+2) = 2^k(k+3)$.

Inductive Step: Using the definition of a_{k+1} , notice that

$$\begin{aligned} a_{k+1} &= 4(a_k - a_{k-1}) \\ &= 4(2^{k-1}(k+2) - 2^{(k-1)-1}((k-1)+2)) && \text{(Inductive Hypothesis)} \\ &= 4(2^{k-1}(k+2) - 2^{k-2}(k+1)) \\ &= 2 \times 2^k(k+2) - 2^k(k+1) \\ &= 2^k(2k+4 - (k+1)) \\ &= 2^k(k+3) \end{aligned}$$

This is what we wanted to prove, so we are done.

MARKING:

1 mark for correct base cases (both required).

1 mark for correct inductive hypothesis, using strong induction (bounds must be correct).

1 mark for correctly identifying what is needed to be shown in the inductive step (either explicitly, or implicitly by a valid inductive step).

1 mark for a valid inductive step.

(2) **(6 marks)** Let

$$A = \{n \in \mathbb{Z}_{>0} \mid n \equiv 1 \pmod{2}, n \leq 7\}, \quad B = \{n \in \mathbb{Z}_{>0} \mid n \equiv 0 \pmod{3}, n < 8\}.$$

Determine the following (no need for justification):

Note that $A = \{1, 3, 5, 7\}$ and $B = \{3, 6\}$.

(a) $A \cap B$

$$\{3\}$$

(b) $A \cup B$

$$\{1, 3, 5, 6, 7\}$$

(c) $A - B$

$$\{1, 5, 7\}$$

(d) $\mathcal{P}(B)$

$$\{\emptyset, \{3\}, \{6\}, \{3, 6\}\}$$

(e) $B \times B$

$$\{(3, 3), (3, 6), (6, 3), (6, 6)\}$$

(f) Number of elements in $A \times B$.

There are 4×2 elements.

MARKING:

1 mark for each part done correctly.